

Ultra Low Specular Reflection LCD with High Visibility for Next Generation Automotive Applications

Takahiro Ishinabe*, Yuichi Kawahira**, Akira Sakai**, Kiyoshi Minoura**

Satoshi Yoshinaga***, Hajime Akimoto***, Taketoshi Nakano***

*Tohoku University, Sendai, Japan

**Sharp Display Technology Corporation, Mie, Japan

***Nichia Corporation, Tokushima, Japan

Abstract

We have developed an ultra-low specular reflection LCD for Next Generation Automotive applications. By combining the FFS-mode LCD with low internal reflection, wide-viewing angle, and no sparkle AGAR film and high-efficiency local dimming mini-LED backlight, we successfully achieved excellent visibility in bright environments and over a wide viewing angle range.

Author Keywords

Low Specular Reflection; Low Internal Reflection; AG Film; Ambient Contrast Ratio; Gloss value; Local Dimming Backlight; Automotive

1. Introduction

In recent years, as displays have become increasingly important in automobiles, there has been a strong demand for large and functional displays, such as pillar-to-pillar displays and curved displays.

Improving visibility is the most critical factor for safety, and it is essential to achieve high visibility regardless of ambient brightness, the angle of incident light, or the viewing angle of the display. Especially as displays become larger, the viewing angles become wider; therefore, improving visibility across a wide-angle range has become important.

Ambient Contrast Ratio (ACR) [1] has been proposed as a method for evaluating image quality in bright environments. It is calculated using the following formula, based on the Specular Component Include (SCI) reflectance measured when the display is off.

$$ACR = \frac{L_w + L_a \cdot R_d}{L_b + L_a \cdot R_d}, \quad L_a = c \cdot \frac{I_a}{\pi}$$

L_w : White luminance of the LCD measured in darkroom
 L_b : Black luminance of the LCD measured in darkroom
 R_d : Reflectance of the LCD
 I_a : Ambient illuminance
 L_a : Ambient luminance
 c : Environment constant

Light reflection on displays, a factor reducing ACR, is classified into specular reflection and diffuse reflection components. Display panels exhibit particularly high specular reflection components. Consequently, techniques to reduce surface reflection and internal reflection within the panel have been proposed as methods to improve ACR.

However, even with reduced panel reflection, strong specular

reflections can significantly decrease the display image's contrast ratio. Furthermore, when strong specular reflections create localized high-brightness regions on the screen, it significantly reduces image visibility due to pupil adjustment. In this case, while contrast ratio remains high in areas without specular reflections, visibility decreases when areas with significantly different luminance or contrast ratios coexist on the screen.

Therefore, suppressing specular reflection is important for improving display visibility. Particularly as displays become larger and viewing angles widen, it is necessary to suppress specular reflection regardless of the angle of ambient light or the viewing angle.

This research reports on the development of a wide-viewing-angle AG film, a technology to suppress specular reflection by optimizing internal reflections within the LCD panel, and a high-brightness local dimming backlight using high-efficiency Mini-LEDs. These innovations have enabled the realization of a display with high image visibility regardless of the viewing environment.

2. Evaluation of Reflective Properties

Environmental factors causing image quality degradation are a decrease in contrast ratio due to specular reflection (glare) and lower uniformity in luminance distribution across the screen. This depends on the display's specular reflection characteristics and is determined by the difference between Specular Component Include (SCI) reflectance and Specular Component Exclude (SCE) reflectance.

However, since SCI and SCE are methods that incident diffuse light onto the panel and measure it from the panel normal direction, they cannot evaluate the angular dependence of the measured object's reflectance characteristics.

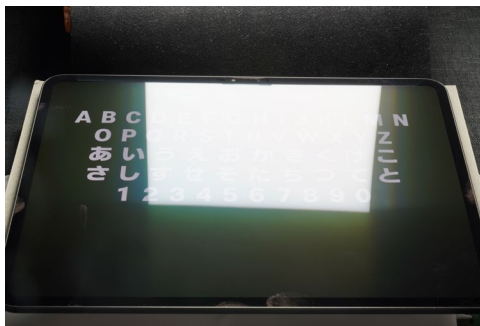
Generally, anti-reflective (AR) coatings and anti-glare (AG) treatments exhibit angle-dependent characteristics, and it is therefore necessary to evaluate their performance for light incident at large angles. In this study, the gloss values at 20°, 60°, and 85° were measured using a gloss meter (GM-268Plus, Konica Minolta) for various display panels. The results are shown in Table 1.

Table 1. Gloss value of various display panels.

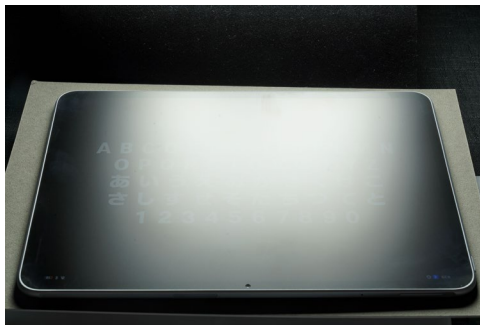
Incident angle	OLED	LCD	LCD+AG
20°	28.6%	37.3%	2.1%
60°	46.8%	58.5%	12.8%
85°	93.1%	94.4%	66.9%

The measurement results show that compared to the glare panel, the anti-glare panel reduces gloss (the intensity of specular reflection) due to light diffusion. However, as the incident angle increases, light diffusion decreases, leading to a higher reflected light intensity. This indicates that to suppress specular reflection (glare) over a wide-angle range, an AG film with high scattering characteristics at large incident angles is necessary.

Next, we evaluated the luminance distribution within the image when specular reflection occurred. A diffuse light-emitting panel was used, illuminating the display at a 45-degree angle and observed from the specular reflection direction. The illuminance was 7,500 lux. For the glare panel, as shown in Fig. 1, light source reflections appear, and the contrast ratio in part of the image is decreased. In contrast, for the anti-glare panel using an AG film, the reflected image of the light source is diffused, resulting in a further decrease in luminance uniformity within the screen. The problem of reduced luminance uniformity in the anti-glare panel is caused by the narrowing of the light diffusion angle due to the angle-dependent characteristics of the AG film's light diffusion properties. This indicates that optimization of the AG film's light diffusion characteristics is necessary.



(a) OLED tablet



(b) LCD tablet with AG

Figure 1. Reflection on various display panels.

3. Development of Wide-viewing-angle AG film

We optimized the average height and average slope of the surface structure of AG film. The characteristics of the fabricated AG film are shown in Table 2.

Figure 2 shows the changes in diffusion characteristics with respect to the incident angle. Solid lines represent the AG film developed in this study, while dashed lines show the characteristics of conventional AG films. The figure indicates that the AG film developed in this study exhibits lower reflected light intensity and has high scattering properties even at large

angles compared to conventional AG films. We also applied an anti-reflective (AR) coating to the fabricated AG film. The SCI reflectance of the fabricated AGAR film is shown in Fig. 3.

Table 1. Optical property of fabricated AG film.

Total Transmittance	90%
Haze	70%

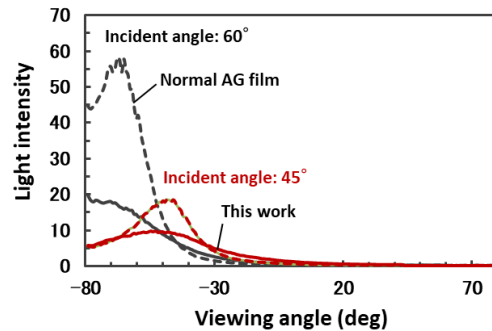


Figure 2. Change in diffusion characteristics of AG film with respect to the incidence angle.

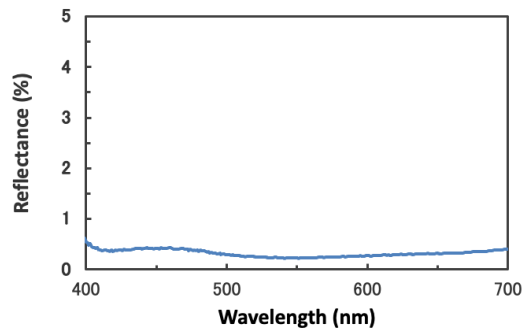
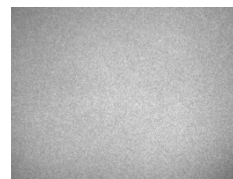
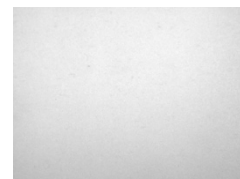


Figure 3. Wavelength - SCI reflectance characteristics of the fabricated AGAR Film.

Here, when AG films have a high haze value, sparkle effects typically occur, reducing image display quality. So, we further developed a phase grating film based on ultraviolet photopolymerization-induced phase separation to reduce sparkle effects. Figure 4 compares the display image (green) from an LCD using conventional AG film without anti-sparkle layers with that from our study. Our results demonstrate that the LCD developed in this study achieves excellent display characteristics with virtually invisible sparkle effects.



(a) LCD with normal AG



(b) LCD with new AG using phase grating film

Figure 4. Comparison of luminance uniformity (sparkle) of LCD panel using AG film.

4. Development of Low-Reflection LCD

We fabricated an LCD with low internal reflection [2-3]. We suppressed refractive index mismatches between ITO and adjacent layers, such as the glass substrate and polarizer adhesive. We also optimized the refractive index of the black resist used for the color filter's BM and the intermediate layer. Furthermore, we optimized the TFT deposition sequence, materials, and layer thicknesses. This resulted in suppression of optical interference between multiple transparent insulating layers, improving the internal reflectance of the panel from the conventional 1.12% to 0.28%.

Additionally, a high-brightness local dimming backlight utilizing mini-LEDs was developed to enhance visibility [4]. Key features include achieving higher brightness and improved luminance uniformity by controlling the directionality of light emission within zones using micro-optics. Furthermore, light leakage into adjacent zones was suppressed, enabling localized light emission while maintaining high contrast. This resulted in image display with minimal halos and excellent contrast. Finally, a high brightness of 3,000 cd/m² was achieved. The specifications of the prototyped LED are shown in Table 3.

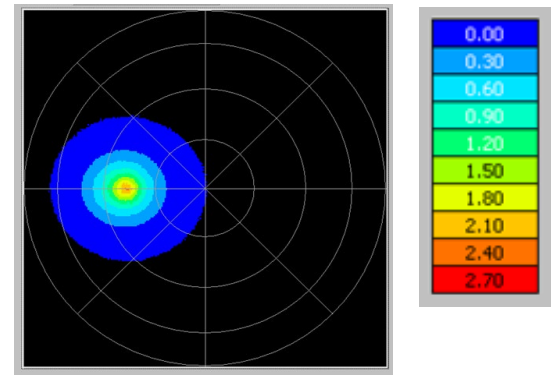
Table 3. Specification of the fabricated LCD.

Display size	7.0inch
Resolution	1280 x 758
Contrast ratio	1630:1
Brightness	3,000 cd/m ²
LCD Panel Internal Reflection	0.28%

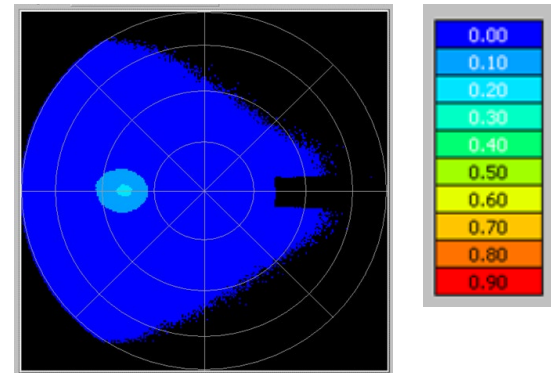
Finally, Table 4 presents a comparison of gloss values for normal LCD+AR, low-reflectance LCD+AR, and low-reflectance LCD+new AGAR. Figure 5 compares the measurement results of the reflection light distribution in the black state measured by the Conoscope (AUTRONIC MELCHERS). The angles of light incidence are 30 degrees and 60 degrees. Figure 6 shows photographs of the displayed images at an illuminance of 7,500 lux. The observation angle is 45 degrees. The LCD developed in this study scatters and reflects external light across a wide angle range, resulting in extremely low glossiness and high luminance uniformity throughout the entire viewing angle range and confirming that specular reflection is effectively suppressed across a wide viewing angle range.

Table 4. Gloss value of various display panels.

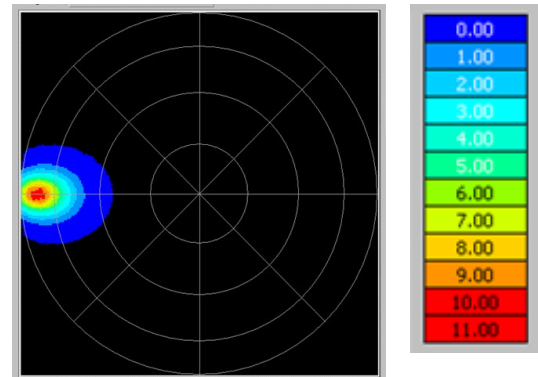
Incident angle	Normal LCD +AR	Low reflection LCD + AR	Low reflection LCD+ New AGAR
20°	17.4%	9.8%	0.2%
60°	50.1%	46.3%	3.0%
85°	95.5%	95.3%	7.5%



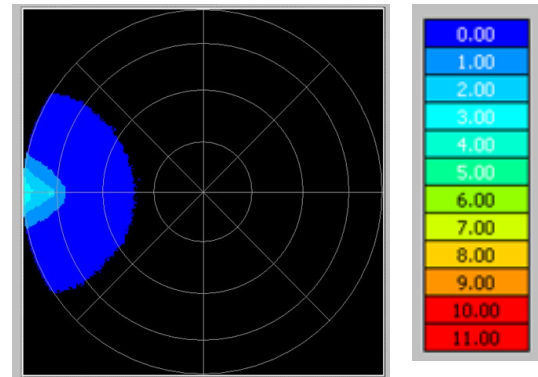
(a) Normal LCD +AG (Incident angle: 30°)



(b) Low reflection LCD + new AG (Incident angle: 30°)



(c) Normal LCD +AG (Incident angle: 60°)



(d) Low reflection LCD + new AG (Incident angle: 60°)

Figure 5. Comparison of the reflection light distribution in the black state.



(a) Normal LCD + AR



(b) Low reflection LCD + New AGAR

Figure 6. Comparison of images under 7,500 lux illumination.

5. Conclusion

In this study, we developed an LCD panel that suppresses internal reflections and a novel AGAR film with extremely low incidence angle dependency and no sparkle effect to improve display visibility in bright environments and over a wide viewing angle range. By combining these technologies with a high-brightness local dimming backlight, we successfully realized an ultra-low specular reflection LCD with excellent visibility that prevents image reflection even under extremely bright conditions of 7,500 lux. This technology is expected to be applied to various applications, including next-generation automotive displays.

6. References

- [1] Haiwei Chen, et al. "Can LCDs Outperform OLED Displays in Ambient Contrast Ratio?" SID'18 DIGEST, 74-1, pp. 981-984 (2018)
- [2] Y. Kawahira, K. Murata, T. Nakai, M. Hasegawa, A. Sakai, K. Minoura, N. Smith, "Sunlight Readable FFS LCD with High Contrast Ratio and Wide Color Gamut under High Ambient Light Situations," Proc. of IDW, pp. 91-94 (2018).
- [3] Y. Kawahira, M. Hasegawa, R. Saigusa, J. Xu, T. Oyama, A. Sakai, K. Minoura, "Super High Ambient Contrast LCDs with Low Power Consumption," SID Display Week Digest, 49-4 (2024).
- [4] R. Yoshimura, T. Nakabayashi, S. Akagawa, "Thin and Low Power 2D-Backlight Module Development with White Mini-LEDs," Proc. of IDW, pp. 337-340 (2023).